Scarce attention and the collapse of joint activity on social networks

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Many of the things we do cannot be done alone. A meeting, a phone call or a shared meal happens only if somebody else chooses it at the same moment. We must decide how much priority to give it without knowing whether the other person will turn up. Queueing models of human activity, which explain the bursty timing of letters and emails [1,2], treat that priority as fixed. We let people choose it [3]. Each person discounts the value of the joint activity by the risk of being stood up, a risk learned from experience; being stood up costs time that could have gone elsewhere.

This single change splits the behaviour into two phases. In one, partners keep meeting. In the other, each has given up on the other, and since nobody offers, nobody finds out that the other would have accepted. The switch between them is abrupt and hard to undo: a pair that has stopped interacting does not restart on its own, and recovery requires one side to keep trying, unanswered, for roughly as long as the other takes to forget. The bursty activity patterns that motivated the queueing models appear only close to this switch.

On a social network the scarce resource is attention. Somebody with many contacts can give each of them only a small share of it, so beyond a critical number of contacts no relationship can be sustained, however valuable. Ties therefore fail first around the best-connected people, and the collapse spreads through the network like a targeted attack on its hubs [4] — except that no attacker is needed, which makes networks with highly connected individuals especially fragile. Simulations show that how many contacts a person has decides which ties fail, while the local structure of the network decides how the failures are arranged (Fig. 1): a network grown by introductions to friends of friends breaks into isolated clusters, while the same network with its links reshuffled loses as many ties but stays in one piece. For activities that need a whole group to be present the limit tightens quickly with group size, giving a largest group that can keep meeting. The theory thus derives, rather than assumes, two facts observed in mobile-phone records: people keep only a limited number of ties active [5], and ties decay when their usual rhythm is interrupted [6].

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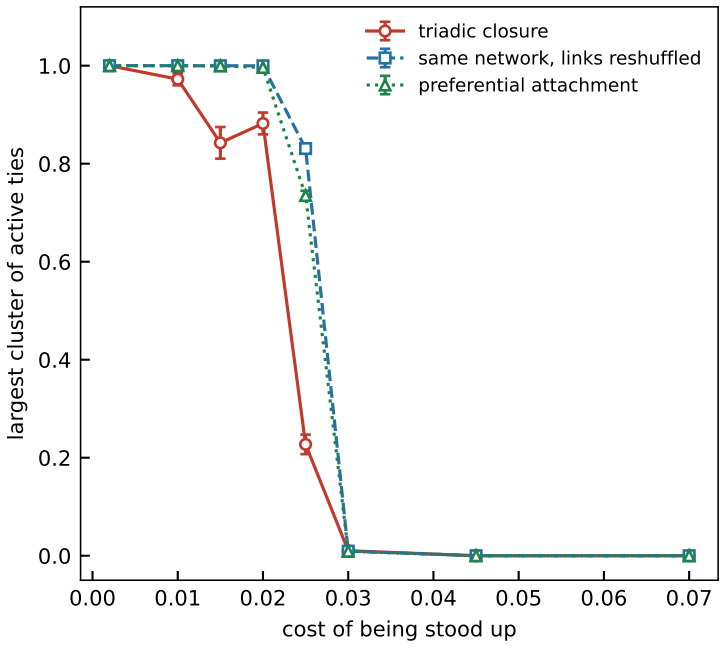


Figure 1: **Local structure decides how joint activity collapses.** Fraction of people in the largest cluster still linked by active ties, as the cost of being stood up grows. A network grown by triadic closure (red) fragments first; the same network with its links reshuffled (blue) keeps the same fraction of active ties but stays connected longer, as does a preferential-attachment network (green). 600 people, 4 contacts on average, 24 simulations per point.